

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Applied Business Analytics Practical

Extra Task

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Use NetFlix.csv

Q.1 Write a Python program
to:

- Find the Top 10 Genres.

Code:

```
# Top 10 Genres

top10 = genre_count.head(10)

print("Top 10 Genres:")

print(top10)
```

Output:

```
Top 10 Genres:
genres
International Movies    2437
Dramas                 2106
Comedies               1471
International TV Shows 1199
Documentaries          786
Action & Adventure     721
TV Dramas              704
Independent Movies     673
Children & Family Movies 532
Romantic Movies        531
Name: count, dtype: int64
```

- Find the Bottom 10 Genres.

Code:

```
# Bottom 10 Genres

bottom10 =
genre_count.tail(10)

print("\nBottom 10 Genres:")

print(bottom10)
```

Output:

```
Bottom 10 Genres:
genres
TV Horror          69
Teen TV Shows      60
Cult Movies        59
Faith & Spirituality 57
Anime Features     57
Movies            56
Stand-Up Comedy & Talk Shows 52
TV Thrillers       50
Classic & Cult TV  27
TV Shows          12
Name: count, dtype: int64
```

- Identify the most popular Genre.

Code:

```
# Most Popular Genre

print("\nMost Popular Genre:")

print(genre_count.idxmax(), "-", genre_count.max())
```

Output:

```
***  
Most Popular Genre:  
International Movies - 2437
```

- Identify the least popular Genre.

Code:

```
# Least Popular Genre  
print("\nLeast Popular  
Genre:")  
  
print(genre_count.idxmin(), "-", genre_count.min())
```

Output:

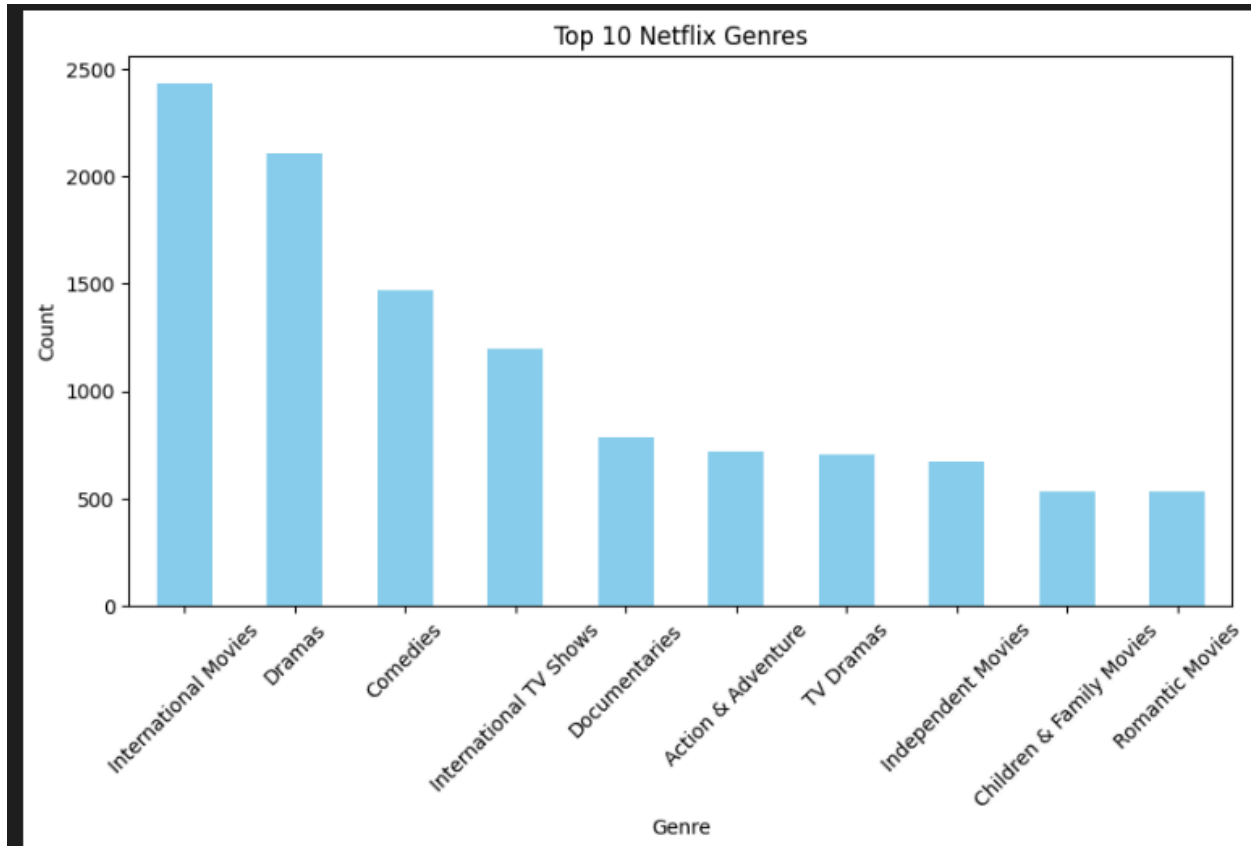
```
Least Popular Genre:  
TV Shows - 12
```

- Display the result using a bar chart.

Code:

```
# Bar Chart  
  
plt.figure(figsize=(10,5))  
  
top10.plot(kind="bar", color="skyblue")  
  
plt.title("Top 10 Netflix Genres")  
  
plt.xlabel("Genre")  
  
plt.ylabel("Count")  
plt.xticks(rotation=45)  
plt.show()
```

Output:



Q.2 Write a Python program to find:

- The oldest Movie.
- The newest Movie.
- The oldest TV Show.
- The newest TV Show.

Display all details of these titles.

Code:

```
# Movies

movies = df[df["type"] == "Movie"]

oldest_movie = movies.loc[movies["release_year"].idxmin()]

newest_movie = movies.loc[movies["release_year"].idxmax()]
```

```
# TV Shows

tv = df[df["type"] == "TV Show"]

oldest_tv = tv.loc[tv["release_year"].idxmin()]
newest_tv = tv.loc[tv["release_year"].idxmax()]

print("Oldest Movie")
print(oldest_movie)

print("\nNewest Movie")
print(newest_movie)

print("\nOldest TV Show")
print(oldest_tv)

print("\nNewest TV Show")

print(newest_tv)
```

Output:

```

... Oldest Movie
  show_id                                s4961
  type                                    Movie
  title                                  Prelude to War
  director                               Frank Capra
  cast                                    NaN
  country                                United States
  date_added                             31-Mar-17
  release_year                           1942
  rating                                 TV-14
  duration                               52
  genres                                Classic Movies, Documentaries
  description    Frank Capra's documentary chronicles the rise ...
  Name: 4402, dtype: object

Newest Movie
  show_id                                s1286
  type                                    Movie
  title                                  Charming
  director                               Ross Venokur
  cast    Wilmer Valderrama, Demi Lovato, Sia, Nia Varda...
  country                                Canada, United States, Cayman Islands
  date_added                             8-Jan-21
  release_year                           2021
  rating                                 TV-Y7
  ...
  duration                               4
  genres                                Kids' TV, TV Thrillers
  description    A master thief who uses her skills for good, C...
  Name: 250, dtype: object

```

Q.3 Write a Python program to:

- Find the country with the highest Netflix content.

Code:

```

# Country with highest Netflix content

print("\nCountry with Highest Netflix
Content:")

print(country_count.idxmax(), "-", country_count.max())

```

Output:

```
Country with Highest Netflix Content:  
United States - 3296
```

- Display the Top 10 Countries.

Code:

```
# Top 10 Countries  
print("\nTop 10  
Countries:")  
  
print(country_count.head(10  
)  
)
```

Output:

```
Top 10 Countries:  
country  
United States      3296  
India               990  
United Kingdom     722  
Canada             412  
France             349  
Japan              287  
Spain              215  
South Korea        212  
Germany            199  
Mexico             154  
Name: count, dtype: int64
```

- Display the Bottom 10 Countries.

Code:

```
# Bottom 10 Countries
```

```
print("\nBottom 10
Countries:")

print(country_count.tail(10))
```

Output:

```
Bottom 10 Countries:
country
Sudan          1
Afghanistan    1
Panama         1
Armenia        1
Mongolia       1
United Kingdom, 1
Uganda         1
East Germany   1
Montenegro     1
Syria          1
Name: count, dtype: int64
```

- Display the result using a horizontal bar chart.

Code:

```
# Horizontal Bar Chart

plt.figure(figsize=(10, 6))

country_count.head(10).sort_values().plot(kind="barh",
color="skyblue")

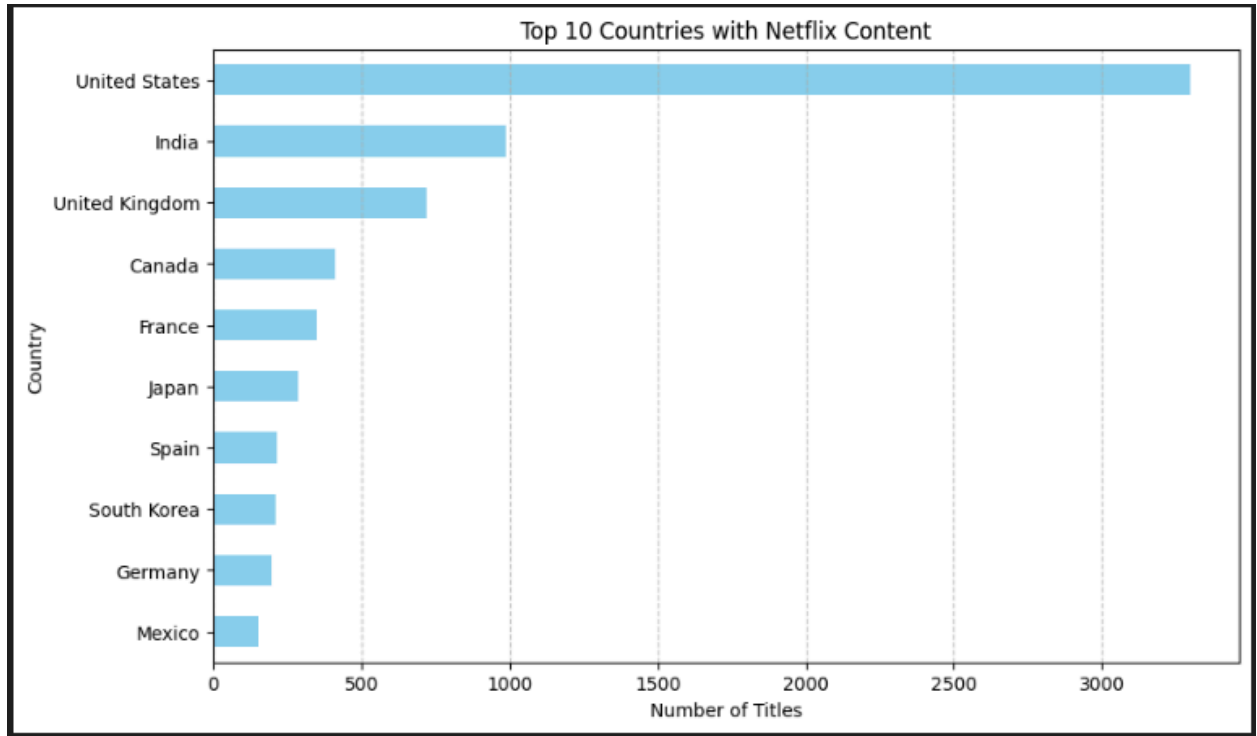
plt.title("Top 10 Countries with Netflix
Content") plt.xlabel("Number of Titles")

plt.ylabel("Country")

plt.grid(axis="x", linestyle="--", alpha=0.7)

plt.show()
```


Output:



Q.4 Write a Python program to:

- Find the director with the highest number of Movies.

Code:

```
# Director with Highest Movies

movie_directors = (

    df[df["type"] == "Movie"]["director"]

    .dropna()

    .str.split(", ")

    .explode()

)

movie_count = movie_directors.value_counts()
```

```
print("\nDirector with Highest Number of Movies:")

print(movie_count.idxmax(), "-", movie_count.max())
```

Output:

```
Director with Highest Number of Movies:
Jan Suter - 21
```

- Find the director with the highest number of TV Shows.

Code:

```
# Director with Highest TV
Shows tv_directors = (

    df[df["type"] == "TV Show"]["director"]

    .dropna()

    .str.split(", ")

    .explode()

)

tv_count = tv_directors.value_counts()

print("\nDirector with Highest Number of TV Shows:")

print(tv_count.idxmax(), "-", tv_count.max())
```

Output:

Director with Highest Number of TV Shows:
Ken Burns - 3

- Display the Top 10 Directors.

Code:

```
# Top 10 Directors (Overall)

top10_directors = (

    df["director"]

    .dropna()

    .str.split(", ")

    .explode()

    .value_counts()

    .head(10)

)

print("\nTop 10 Directors:")

print(top10_directors)
```

Output:

```

Top 10 Directors:
director
Jan Suter                21
Raúl Campos              19
Marcus Raboy             16
Jay Karas                15
Cathy Garcia-Molina      13
Youssef Chahine          12
Jay Chapman              12
Martin Scorsese          12
Steven Spielberg         10
Shannon Hartman           9
Name: count, dtype: int64

```

- Plot a bar chart of the Top 10 Directors.

Code:

```

# Bar Chart

plt.figure(figsize=(10,6))

top10_directors.plot(kind="bar", color="green")

plt.title("Top 10 Netflix
Directors") plt.xlabel("Director")

plt.ylabel("Number of Titles")

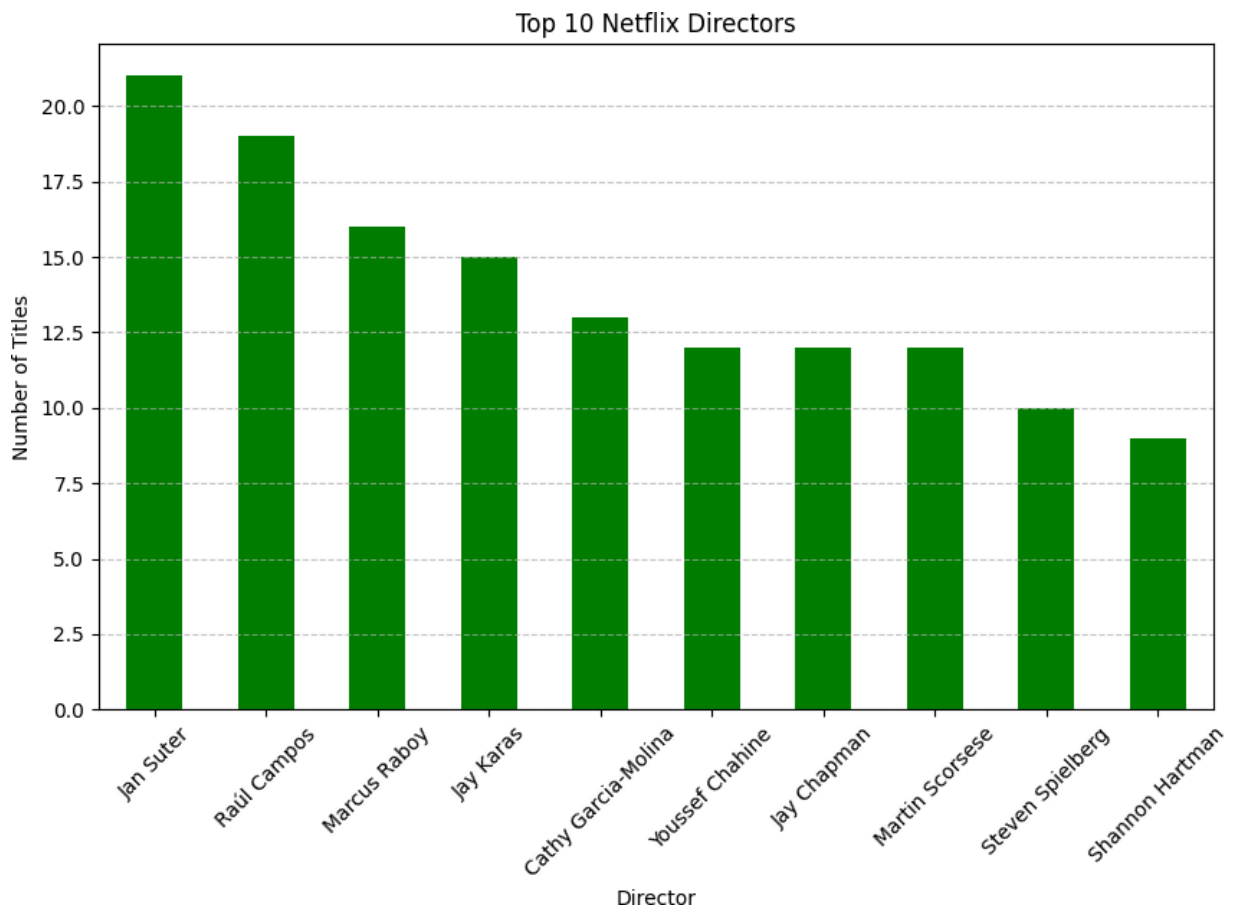
plt.xticks(rotation=45)

plt.grid(axis="y", linestyle="--", alpha=0.7)

plt.show()

```

Output:



Q.5 Compare Netflix releases:

- Before 2020.
- During 2020.
- After 2020.
- Display the result using a bar chart or line chart.

Code:

```
before = len(df[df["release_year"] < 2020])  
  
during = len(df[df["release_year"] == 2020])  
  
after = len(df[df["release_year"] > 2020])
```

```
print("Before 2020 :",
before) print("During 2020
:", during) print("After
2020 :", after)

years = ["Before 2020", "During 2020", "After 2020"]
counts = [before, during, after]

plt.figure(figsize=(6,5))
plt.bar(years, counts, color=["blue", "red",
"green"]) plt.title("Netflix Releases Comparison")
plt.xlabel("Year Category")

plt.ylabel("Number of Titles")

plt.show()
```

Output:

Before 2020 : 6888

During 2020 : 868

After 2020 : 31

